EE2015: Electric Circuits and Networks

Tutorial 10 Answer Key
(December 10, 2025)

1. $\mathbf{V}_{an} = 200\angle 0°\,\text{V}$, $\mathbf{V}_{bn} = 200\angle -120°\,\text{V}$, $\mathbf{V}_{cn} = 200\angle 120°\,\text{V}$
   $\mathbf{I}_{aA} = 2\angle -60°\,\text{A}$, $\mathbf{I}_{bB} = 2\angle -180°\,\text{A}$, $\mathbf{I}_{cC} = 2\angle 60°\,\text{A}$

   $\mathbf{V}_{ab} = 346\angle 30°\,\text{V}$, $\mathbf{V}_{bc} = 346\angle -90°\,\text{V}$, $\mathbf{V}_{ca} = 346\angle 150°\,\text{V}$
   $\mathbf{I}_{aA} = 2\angle -60°\,\text{A}$, $\mathbf{I}_{bB} = 2\angle -180°\,\text{A}$, $\mathbf{I}_{cC} = 2\angle 60°\,\text{A}$

   $P_{aA} = 200\,\text{W}$, Total power $= 600\,\text{W}$

2. (a) $\mathbf{V}_{an} = 231\angle -30°\,\text{V}$, (b) $\mathbf{I}_{aA} = 4.62\angle -30°\,\text{A}$, (c) Total power $= 3200\,\text{W}$

3. $2.89\angle -36.9°\text{A}$

4. (a) $\mathbf{I}_{AB}^{(1)} = 30.3\angle -6.87°\,\text{A}$, (b) $\mathbf{I}_{AB}^{(2)} = 20.2\angle 55.8°\,\text{A}$, (c) $\mathbf{I}_{aA} = 75.3\angle -12.46°\,\text{A}$
   (d) Total power: $56.0\,\text{kW}$

5. $\mathbf{Z}_{\text{ph}} = 407.742\angle 31.78°\,\Omega$, $\mathbf{I}_{\text{line}} = 1.69\angle -1.78°\,\text{A}$

   $\mathbf{Z}_{\text{ph}} = 1471.9\angle -23.09°\,\Omega$, $\mathbf{I}_{\text{line}} = 0.47\angle 53.09°\,\text{A}$

6. (a) $980\,\text{W}$, (b) $988\,\text{W}$, (c) $195.1\,\text{W}$, (d) $720\,\text{W}$, (e) $692\,\text{W}$

7. (a) Use $\mathbf{Z}_L/a^2$ to find the reflected impedance on the primary side
   (b) What is the factor that is required to transfer the primary side impedance to the secondary side? Which other parameter uses the turns ratio?